Introduction to PyTorch

What is PyTorch?

- PyTorch is an **open-source** deep learning framework developed by Facebook's AI Research lab (FAIR) https://pytorch.org/

- PyTorch provides a **flexible** and efficient platform for building and training neural networks.

- It was first released in 2016 and has since become **one of the most widely used frameworks in academia and industry**.

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What PyTorch helps with?

- Explicit code, flexibility
- Large community
- Production-ready deployment
- Domain-specific extensions : TorchVision, TorchText, TorchAudio
- Integration with Hugging Face Transformers for NLP and generative AI

Hugging Face

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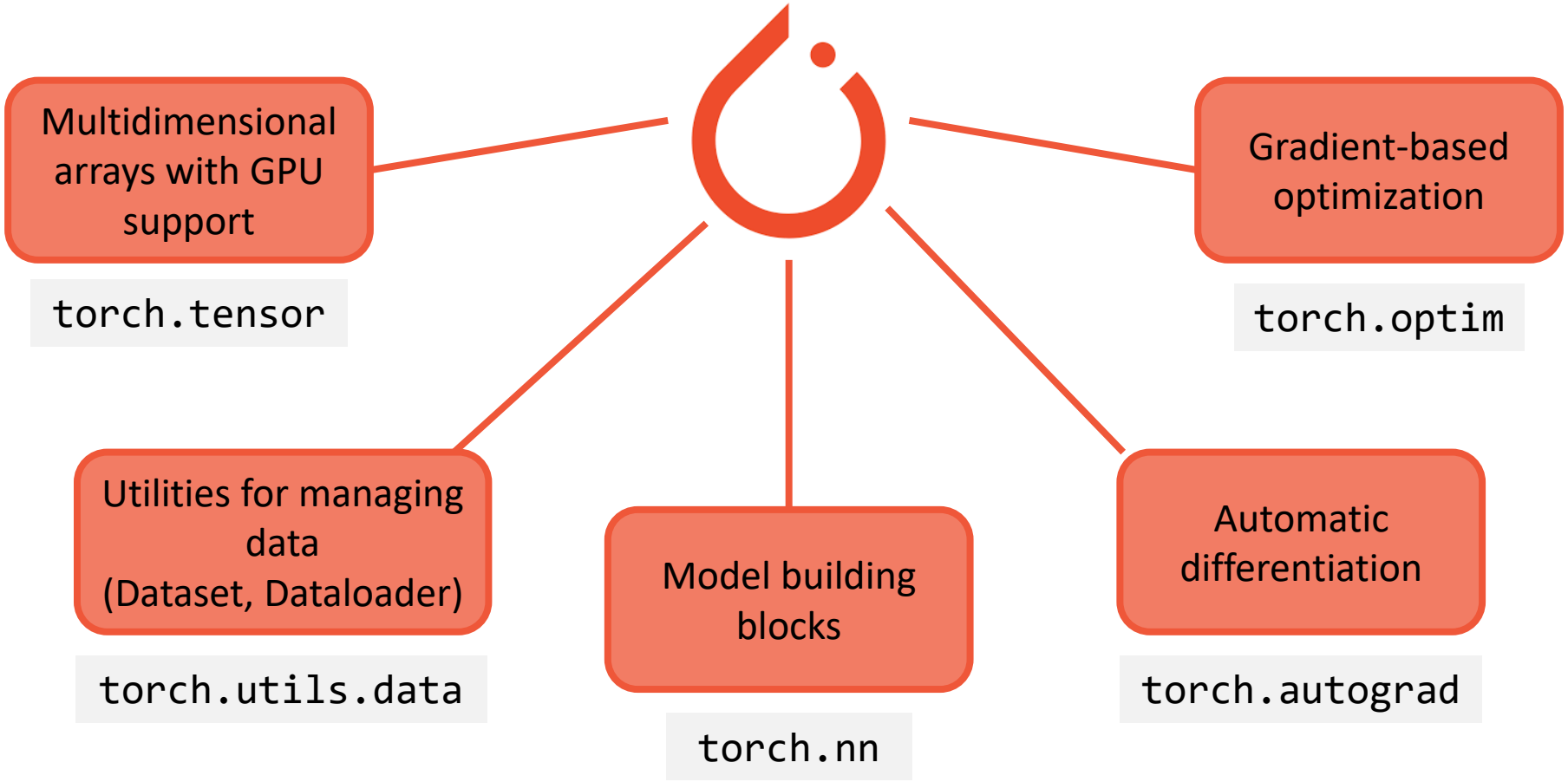
What PyTorch helps with?



Other libraries

NumPy: Scientific computing package
https://numpy.org/

Scikit-Learn: Classical machine learning algorithms and preprocessing. Built on NumPy, SciPy, and Matplotlib
https://scikit-learn.org/stable/

Matplotlib: Visualization library for static plots. https://matplotlib.org/

Seaborn: high level library, built on matplotlib.
https://seaborn.pydata.org/

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What is Google Colab?

- Google Colab is a web IDE (Integrated Development Environment) for Python https://colab.research.google.com/
- Jupyter Notebook environment directly in the browser, connected to cloud servers
- Environment is already setup, with popular ML/DL packages (PyTorch, TensorFlow, scikit-learn, pandas, etc.)
- Unlimited access to CPU, limited but free access to GPU (Nvidia T4)
- Enables access to Google Drive 

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Time to practice!

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Common Data Science Datasets

- MNIST: Handwritten digits
- [PyTorch Datasets]

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Common Data Science Datasets

- Fashion-MNIST: Zalando fashion item classification
- PyTorch Datasets

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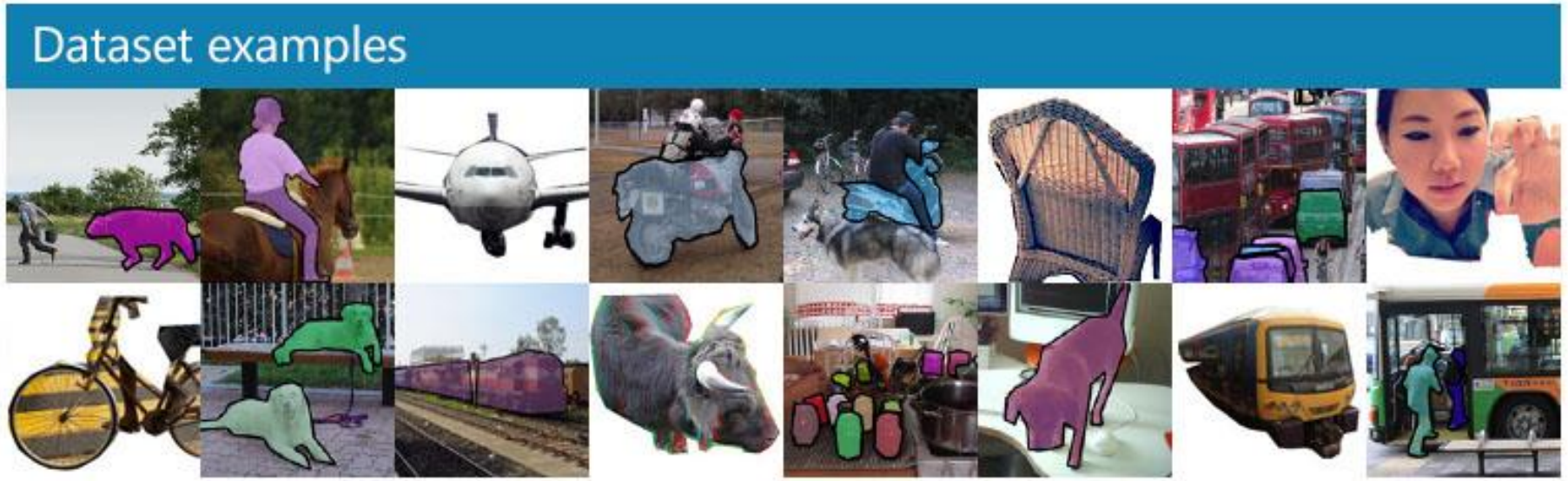
Common Data Science Datasets

- COCO: : Common Objects in COntext
- PyTorch Datasets



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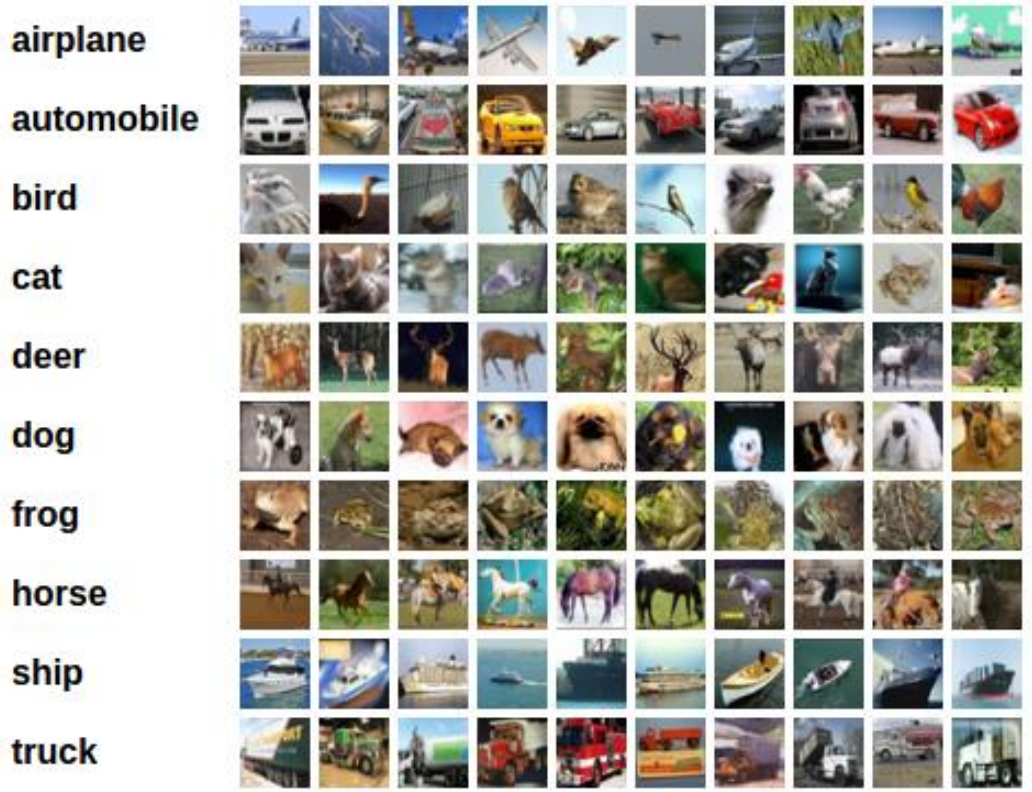
Common Data Science Datasets

- CIFAR-10 / CIFAR-100: Object classification in small images
- PyTorch Datasets



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Common Data Science Datasets

- ImageNet: Large-scale visual recognition
- PyTorch Datasets

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Common Data Science Datasets

- Wine dataset: Wine quality classification
- Scikit-Learn Datasets

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	9.8	5
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	9.8	5
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	6
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5

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Common Data Science Datasets

- Amazon Review Dataset: Huge dataset for sentiment analysis and recommendation systems
- Hugging Face Datasets

	Product Name	Brand Name	Price	Rating	Reviews	Review Votes
82379	ASUS ZenFone 3 MAX 5.2-inch (2GB RAM, 16GB sto...	NaN	149.99	5	Faster, good quality camera and the fingerprin...	6.0
232360	LG Nexus 5X Unlocked Smartphone with 5.2-Inch ...	LG	399.00	5	Love the cell very good one in every thing	0.0
39160	Apple iPhone 5c Factory Unlocked Cellphone, 8G...	NaN	199.99	5	The Phone came unlocked & its a sweet surprise...	0.0
166311	CHSLING Unlocked Android MTK6580 Quad Core ROM...	NaN	79.99	3	Its fairly ok, but really not a high end phone...	0.0
231945	LG Nexus 5X Unlocked Smartphone - White 32GB (...	LG	399.00	1	Thus phone worked great for about 3 months. Th...	7.0
30001	Apple iPhone 5c 32GB (Blue) - AT&T	Apple	274.95	5	What an upgrade compared to the iPhone 4. Goin...	7.0
313198	Samsung Galaxy Grand Prime DUOS G531H/DS - Gra...	Samsung	179.99	4	I liked it at first but is starting to lag alr...	0.0
138219	BLU Studio 5.0 C HD Unlocked Cellphone, White	BLU	2000.00	4	very nice	0.0
66571	Apple iPhone 6s 64 GB International Warranty U...	Apple	689.95	1	It is not a new one. The tagboard on the box w...	0.0
109303	BLU Dash J Unlocked Phone - Retail Packaging -...	BLU	39.99	1	This phone was truly a terrible purchase!! It ...	1.0

Where to find datasets for free?

- PyTorch Datasets
- Keras Datasets (only 7)
- Scikit-Learn Datasets
- Kaggle Datasets
- Hugging Face Datasets
- Google Dataset Search

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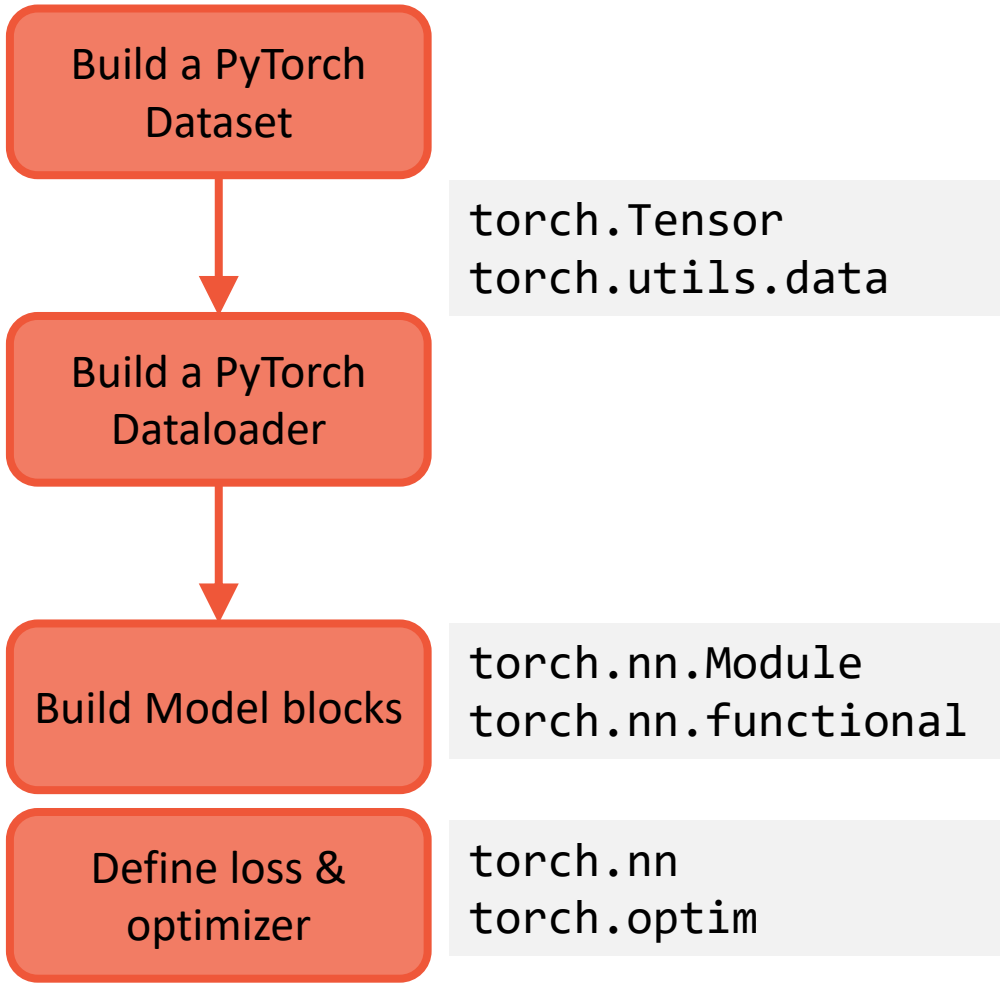
PyTorch data and model setup



- The Dataset object transforms data in a format that PyTorch can handle
- Dataloader object determines how to feed the model with data (mini batch, shuffling or not, etc.)
- Model object defines structure: layers, activation functions, etc.
- Criterion & optimizer define how loss and parameter optimization is computed

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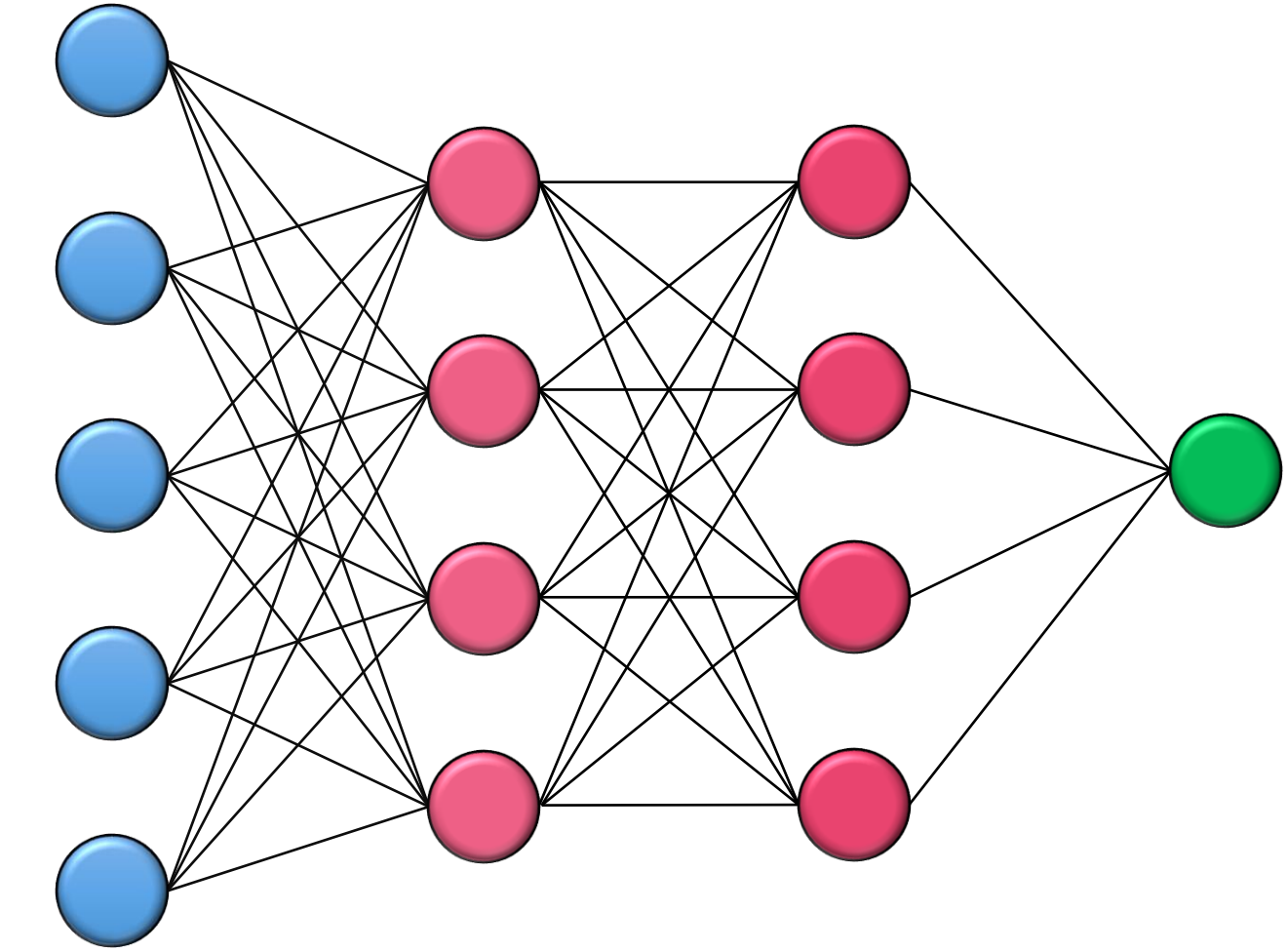
Neural network representation: layers & connexions



input layer hidden layer 1 hidden layer 2 output layer

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Neural networks are essentially matrix multiplications + activations

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Input vector

$$W = \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{13} \\ w_{31} & w_{32} & w_{13} \\ w_{41} & w_{42} & w_{43} \end{bmatrix} \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

Weight matrix Bias vector

One hidden layer

Linear relationship

$$Z = WX + b = \begin{bmatrix} w_{11}x_1 + w_{12}x_2 + w_{13}x_3 + b_1 \\ w_{21}x_1 + w_{22}x_2 + w_{23}x_3 + b_2 \\ w_{31}x_1 + w_{32}x_2 + w_{33}x_3 + b_3 \\ w_{41}x_1 + w_{42}x_2 + w_{43}x_3 + b_4 \end{bmatrix}$$

Activation (layer output)

$$A = f(Z) = [z_1 \quad z_2 \quad z_3]$$

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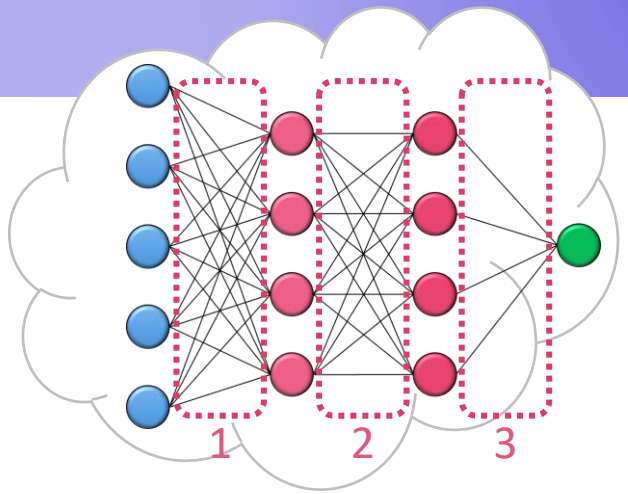
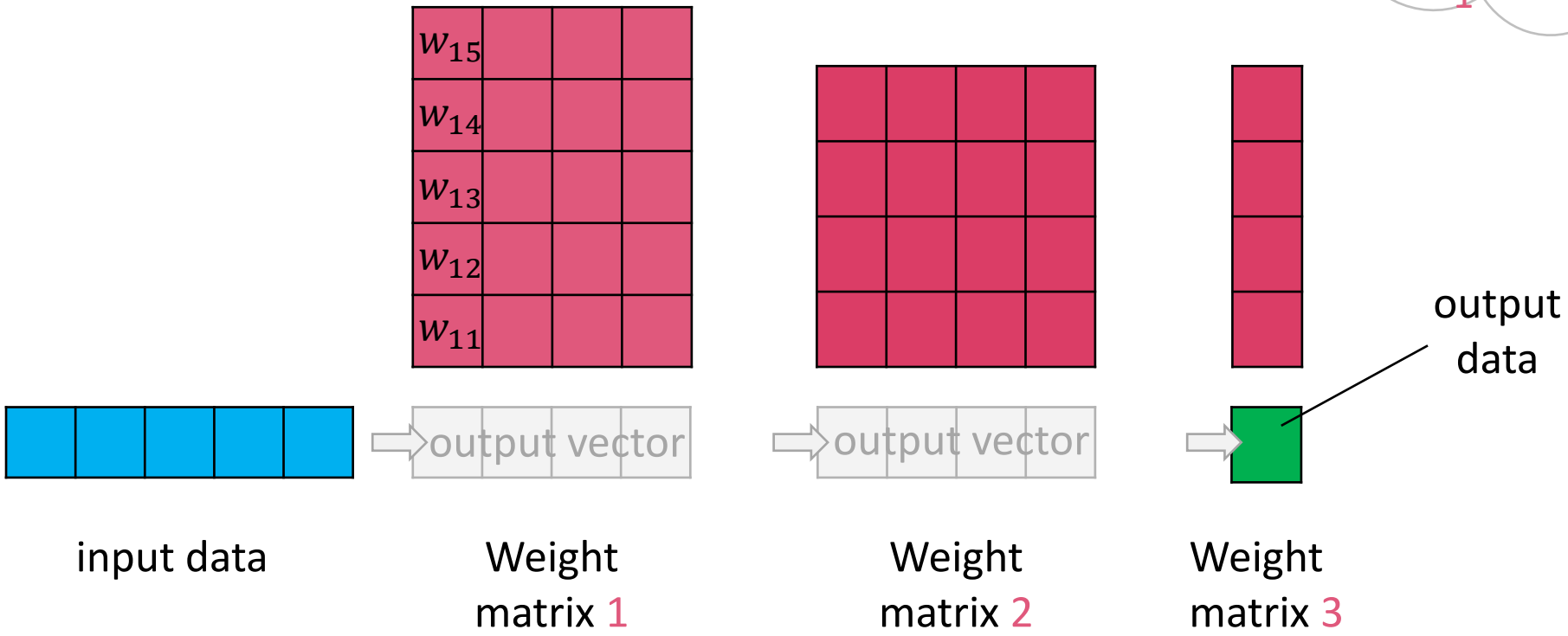
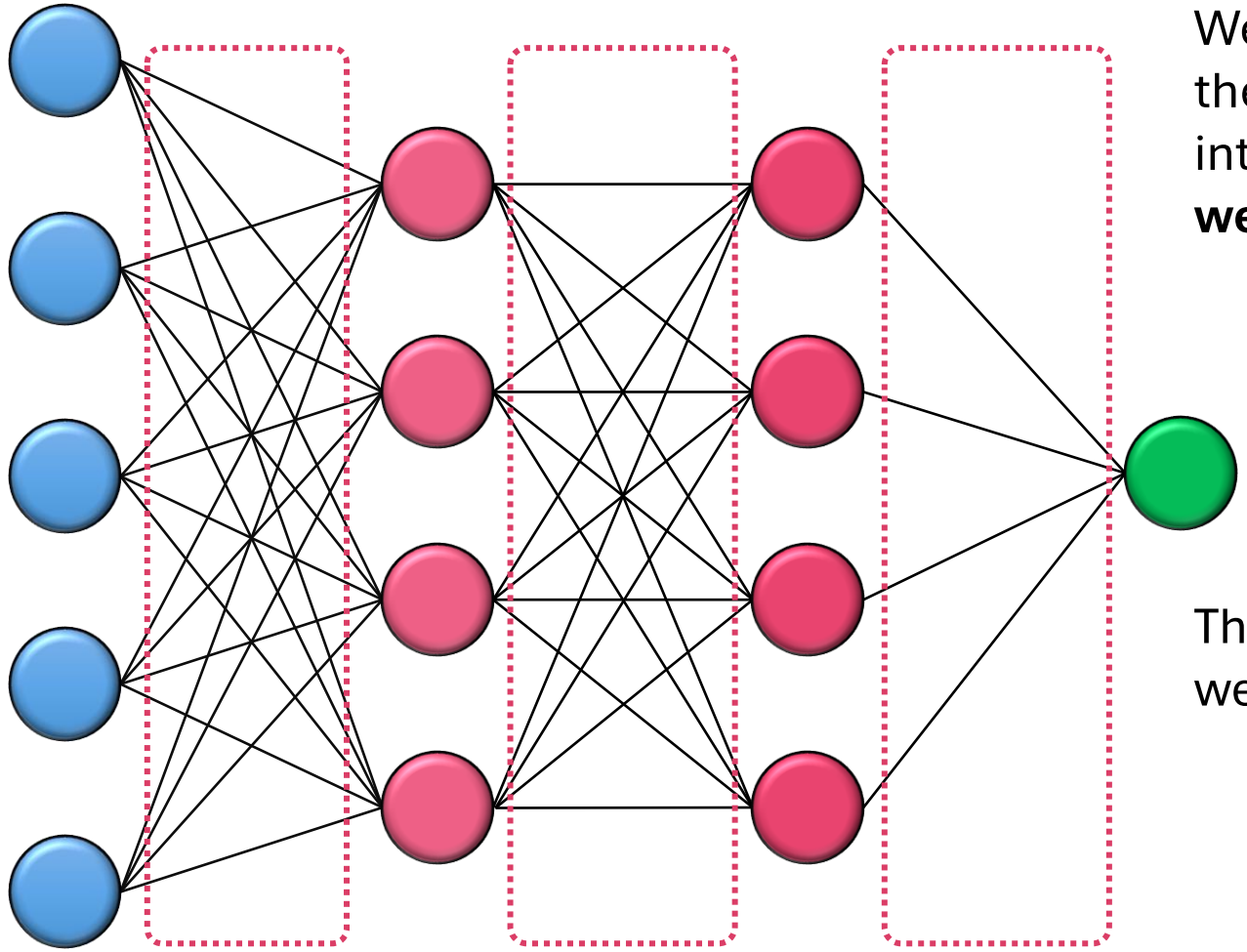
Neural network representation: vectors & matrices

- The same neural network can be represented like this
- Better understanding of calculations and shapes





How to build a model class with PyTorch



`torch.nn.Linear(5, 4)`

`torch.nn.Linear(4, 4)`

`torch.nn.Linear(4, 1)`

We are going to put each of these fully connected layers into layer objects which are **weight matrices**

The model will learn these weight matrices

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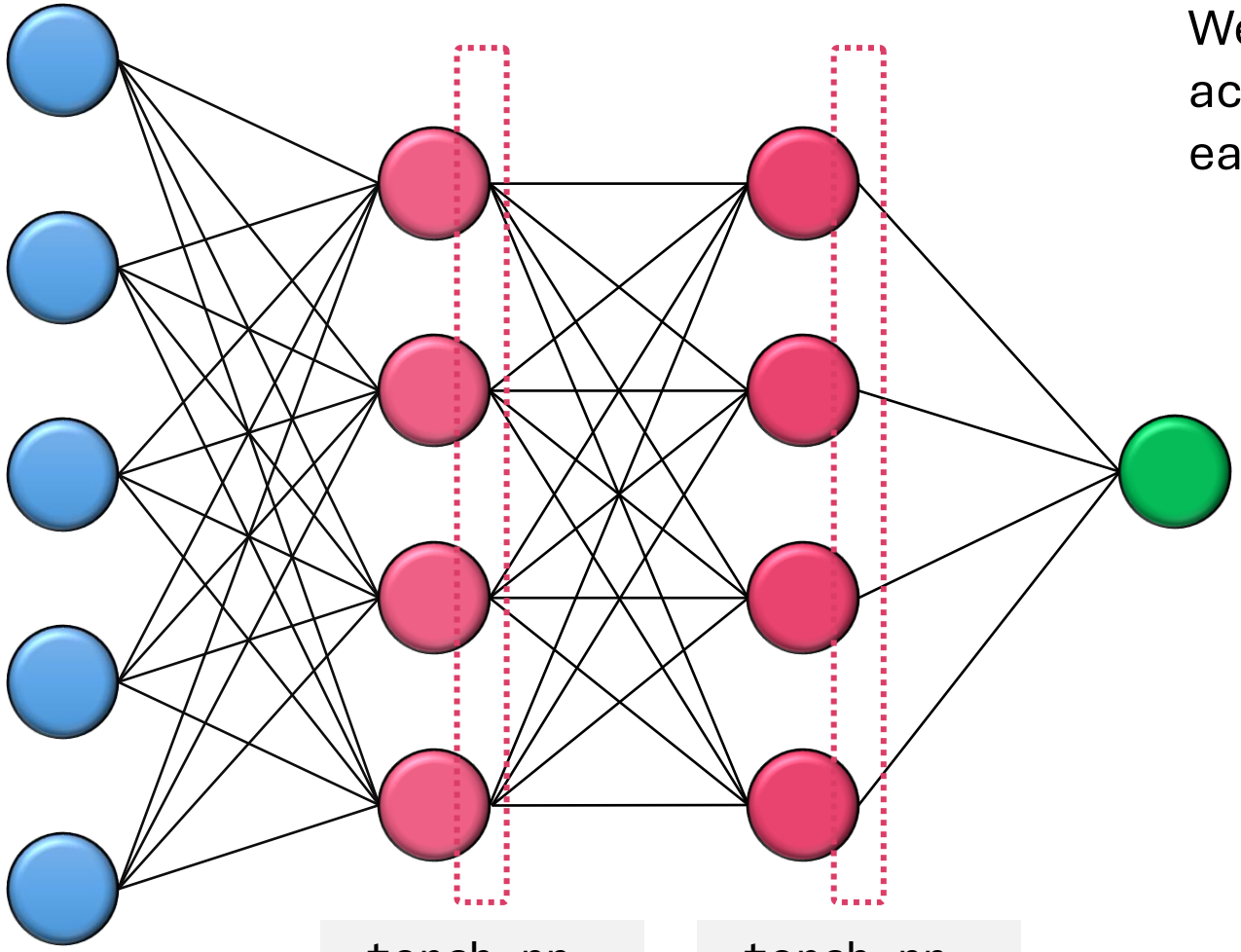
How to build a model class with PyTorch

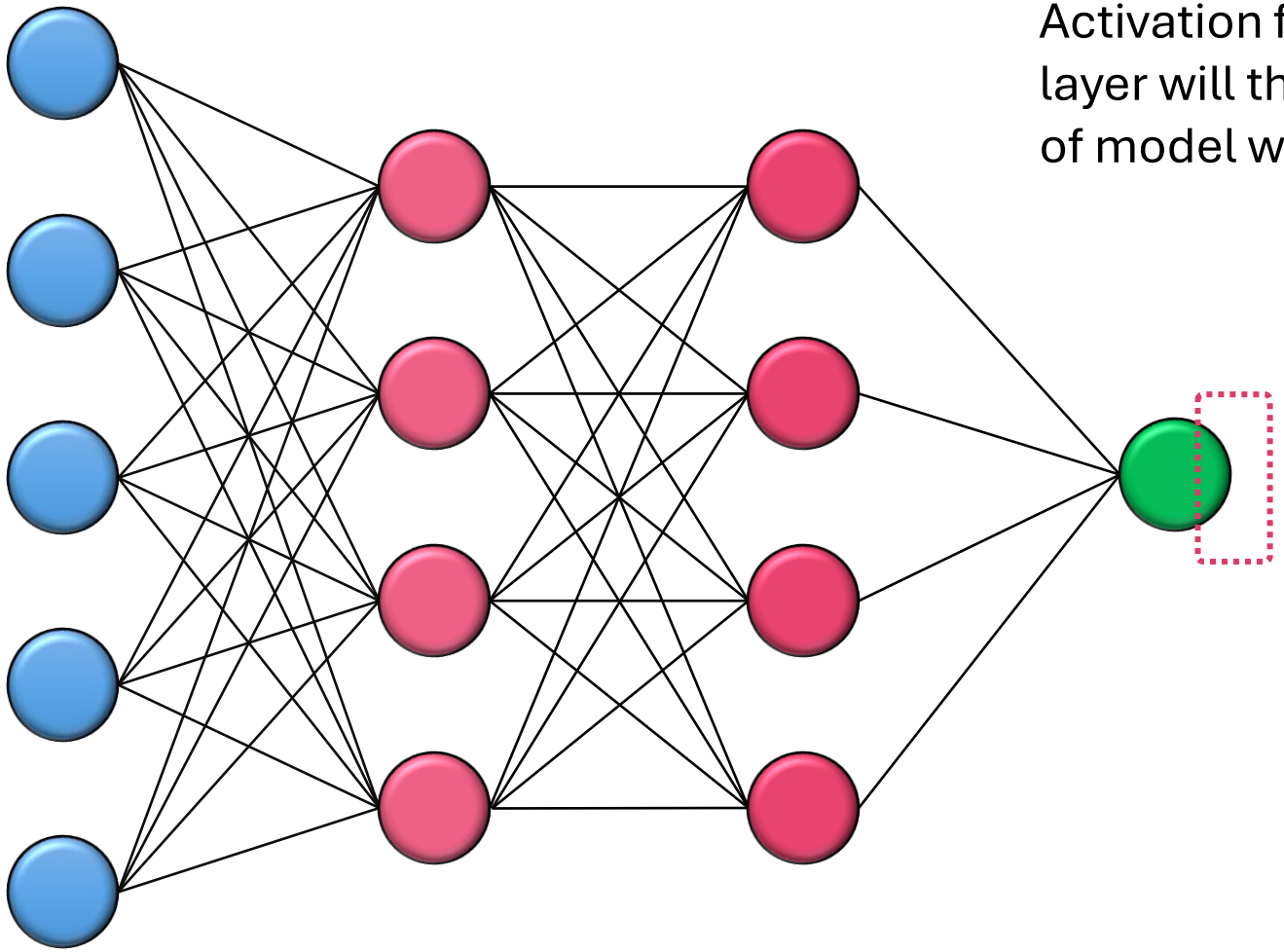


`torch.nn.Functional.relu()`

`torch.nn.Functional.relu()`

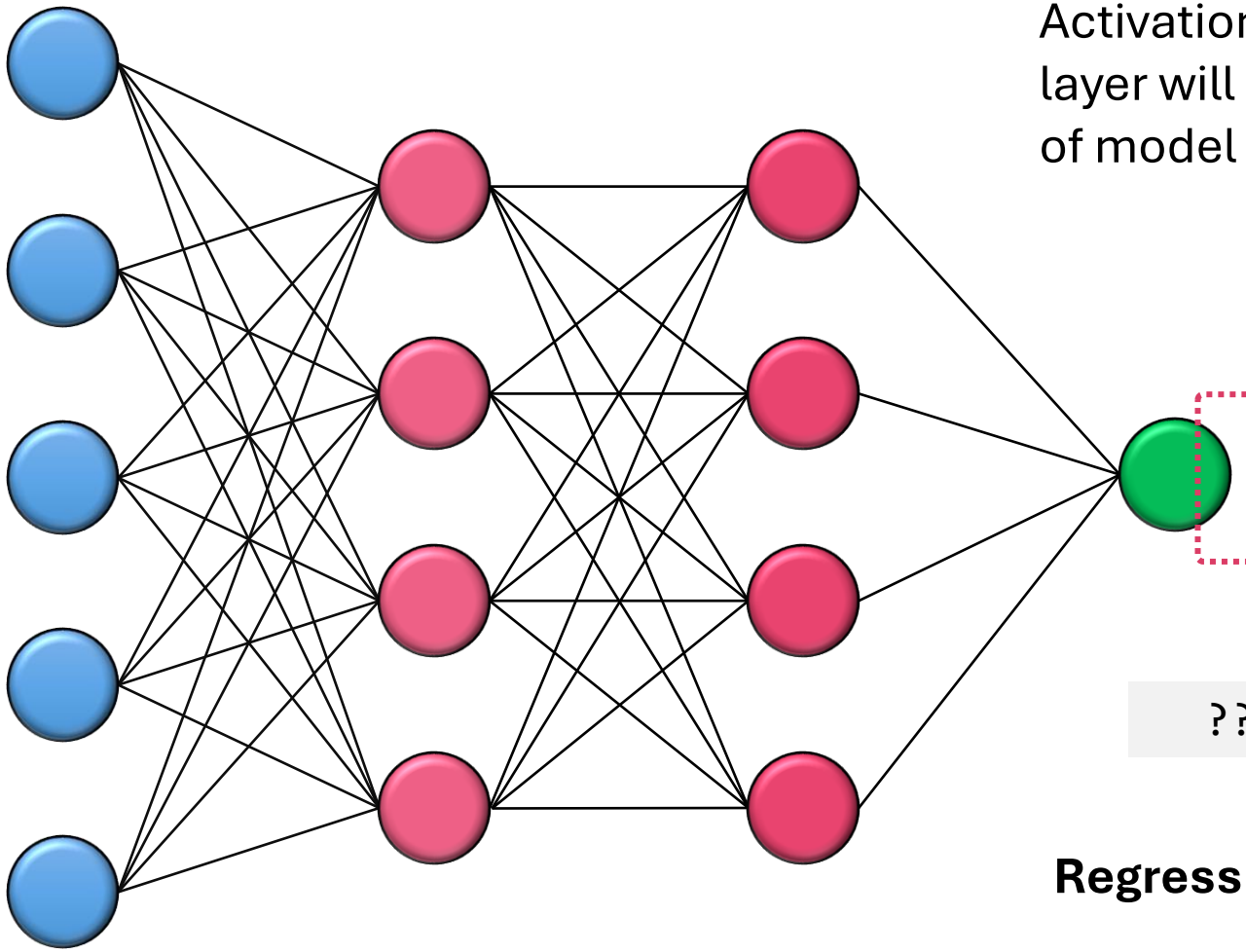
We are also going to define activation functions for each node output

How to build a model class with PyTorch



Activation function for the output layer will then define which type of model we want

How to build a model class with PyTorch

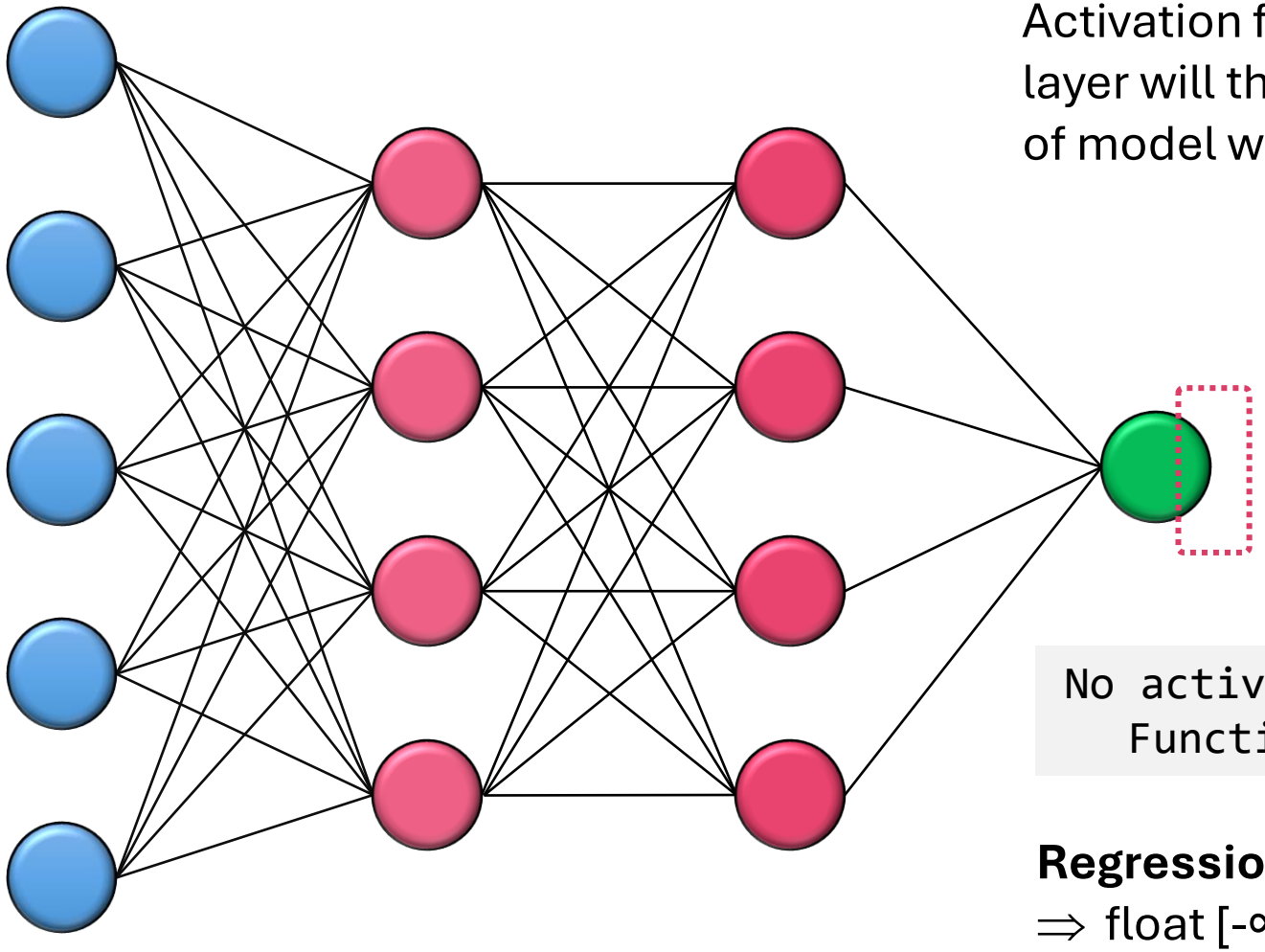


Activation function for the output layer will then define which type of model we want

??

Regression model

How to build a model class with PyTorch



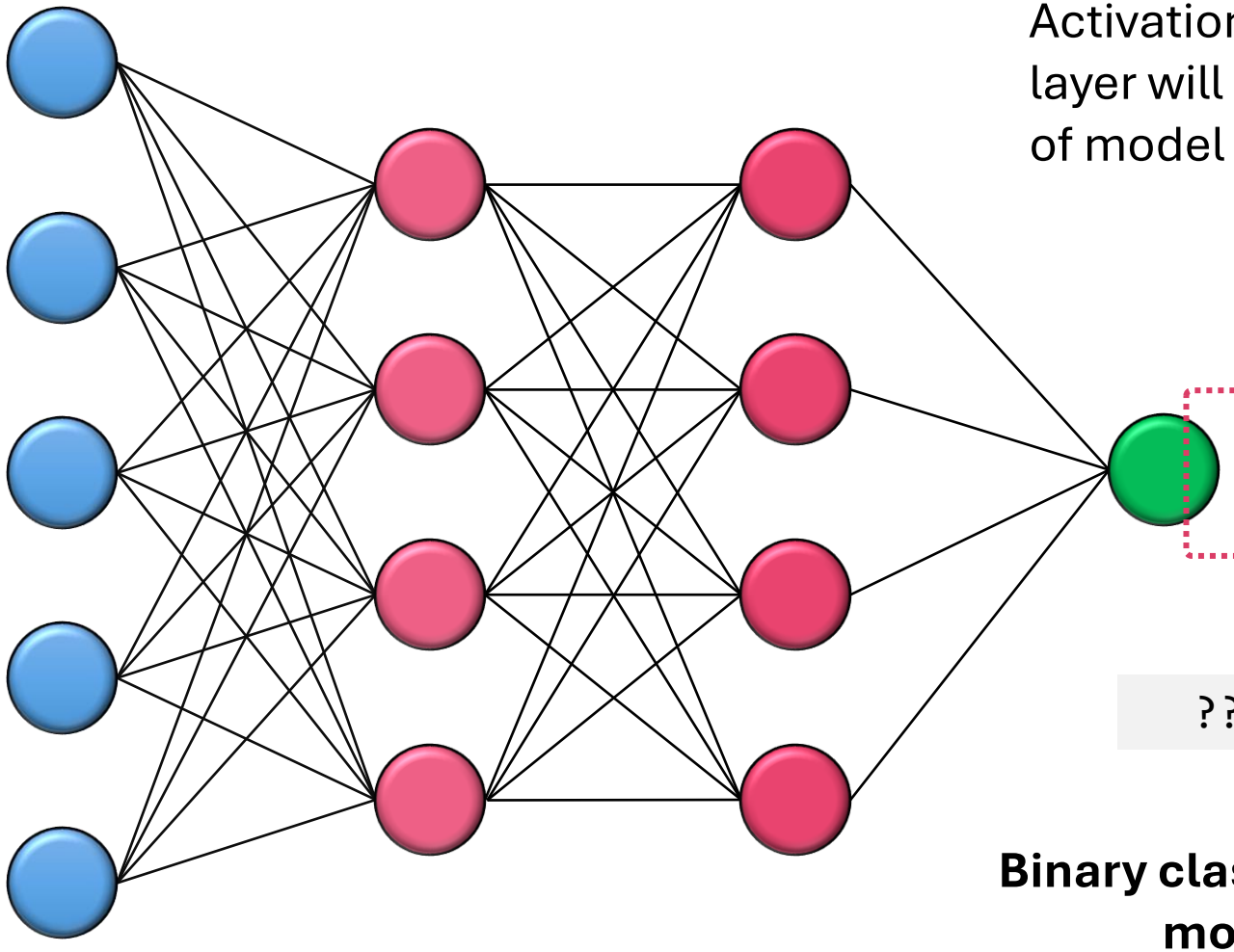
Activation function for the output layer will then define which type of model we want

```
No activation
Function
```

Regression model

$\Rightarrow$ float [-∞; +∞]

How to build a model class with PyTorch

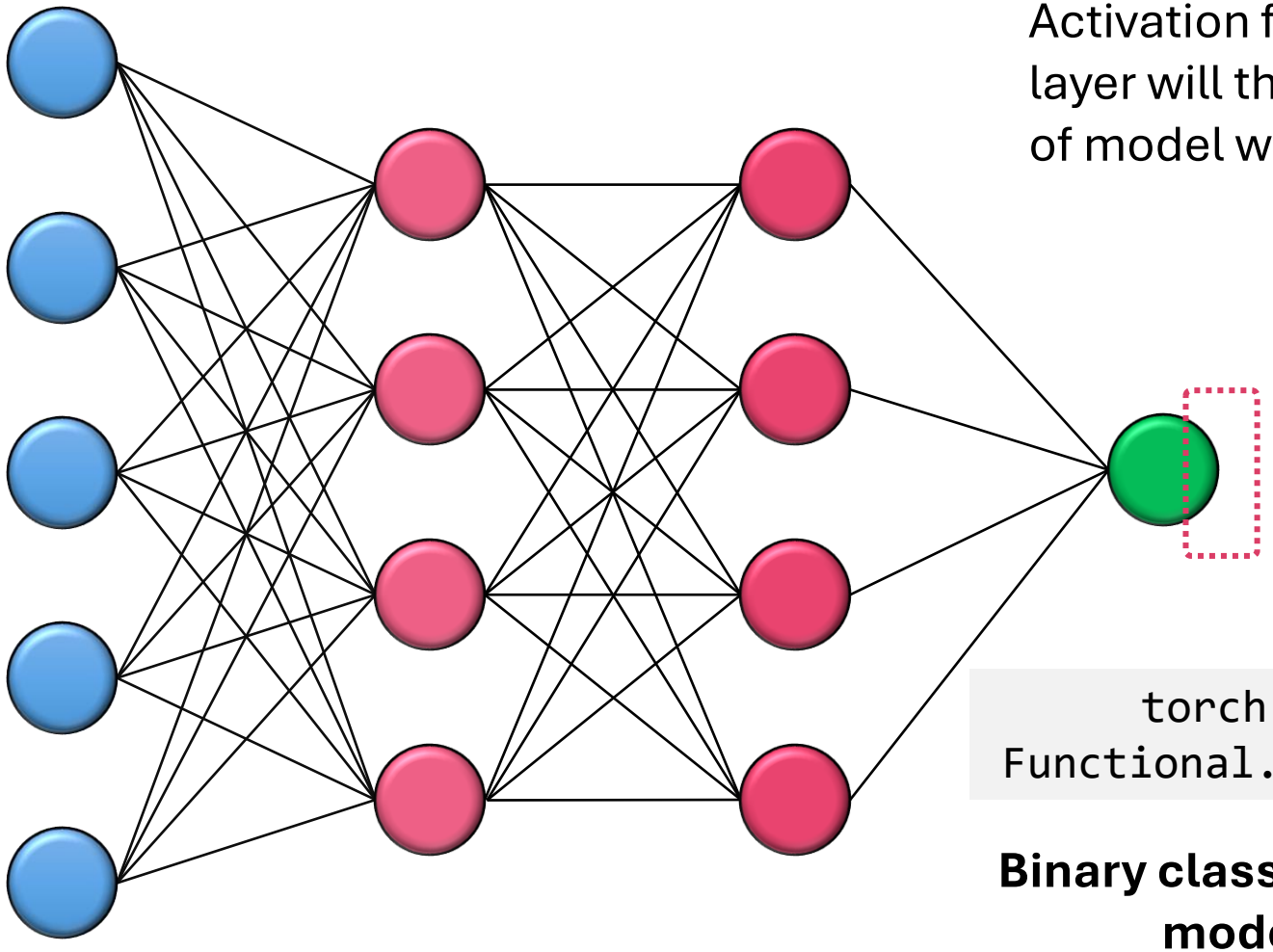


Activation function for the output layer will then define which type of model we want

??

Binary classification model

How to build a model class with PyTorch



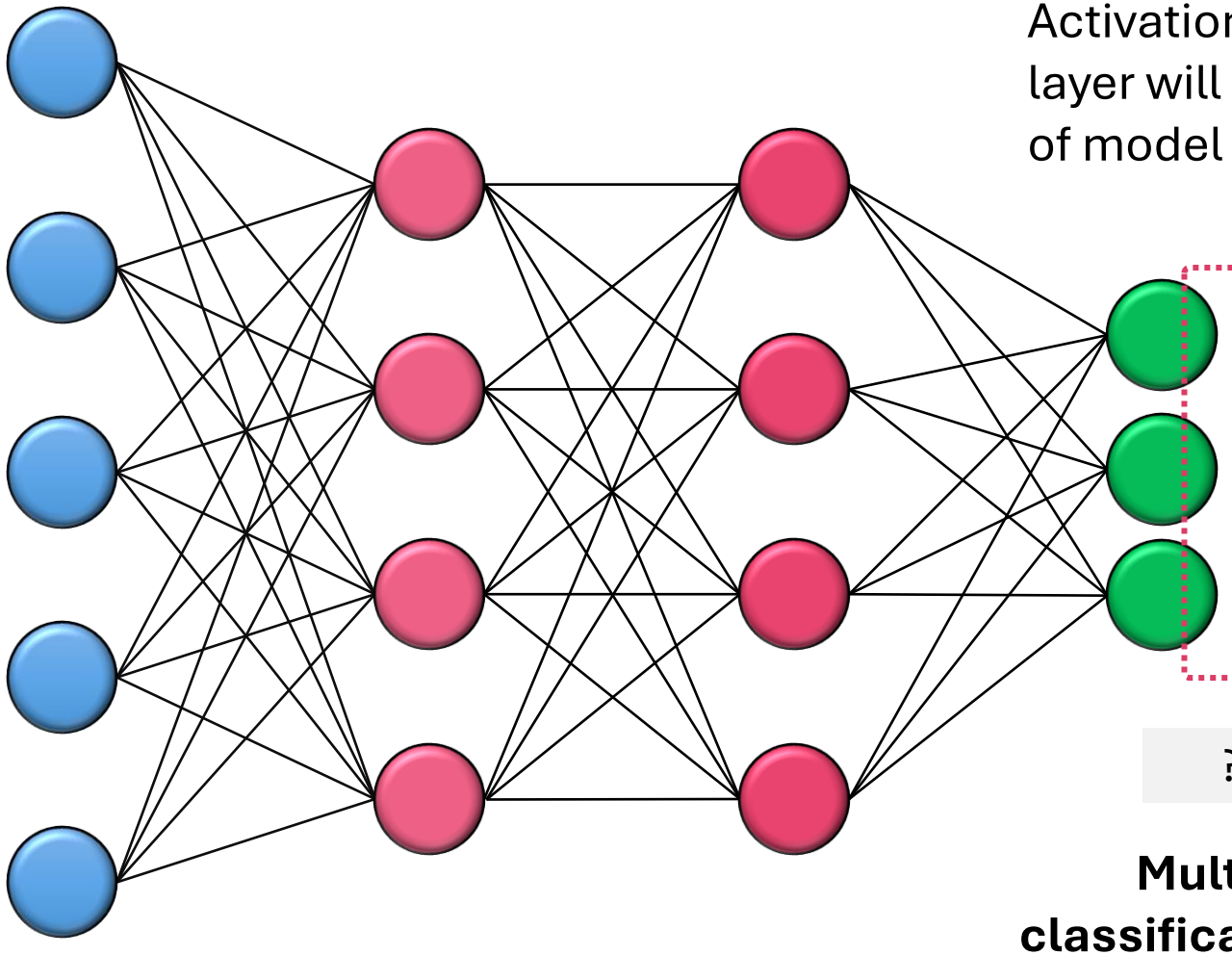
Activation function for the output layer will then define which type of model we want

```
torch.nn.
Functional.sigmoid()
```

Binary classification model

$\Rightarrow$ probability [0; 1]

How to build a model class with PyTorch

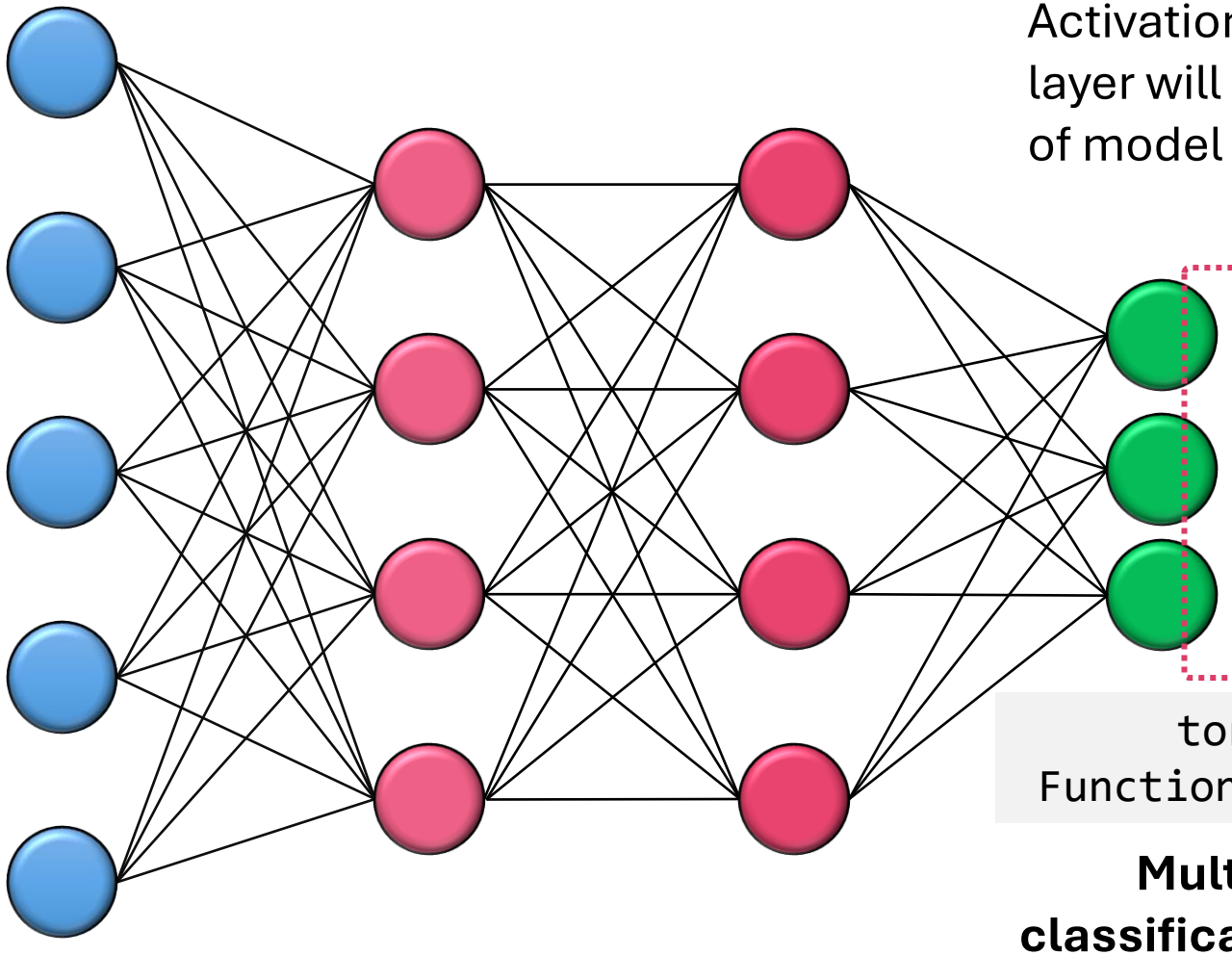


Activation function for the output layer will then define which type of model we want

??

Multiclass classification model

How to build a model class with PyTorch



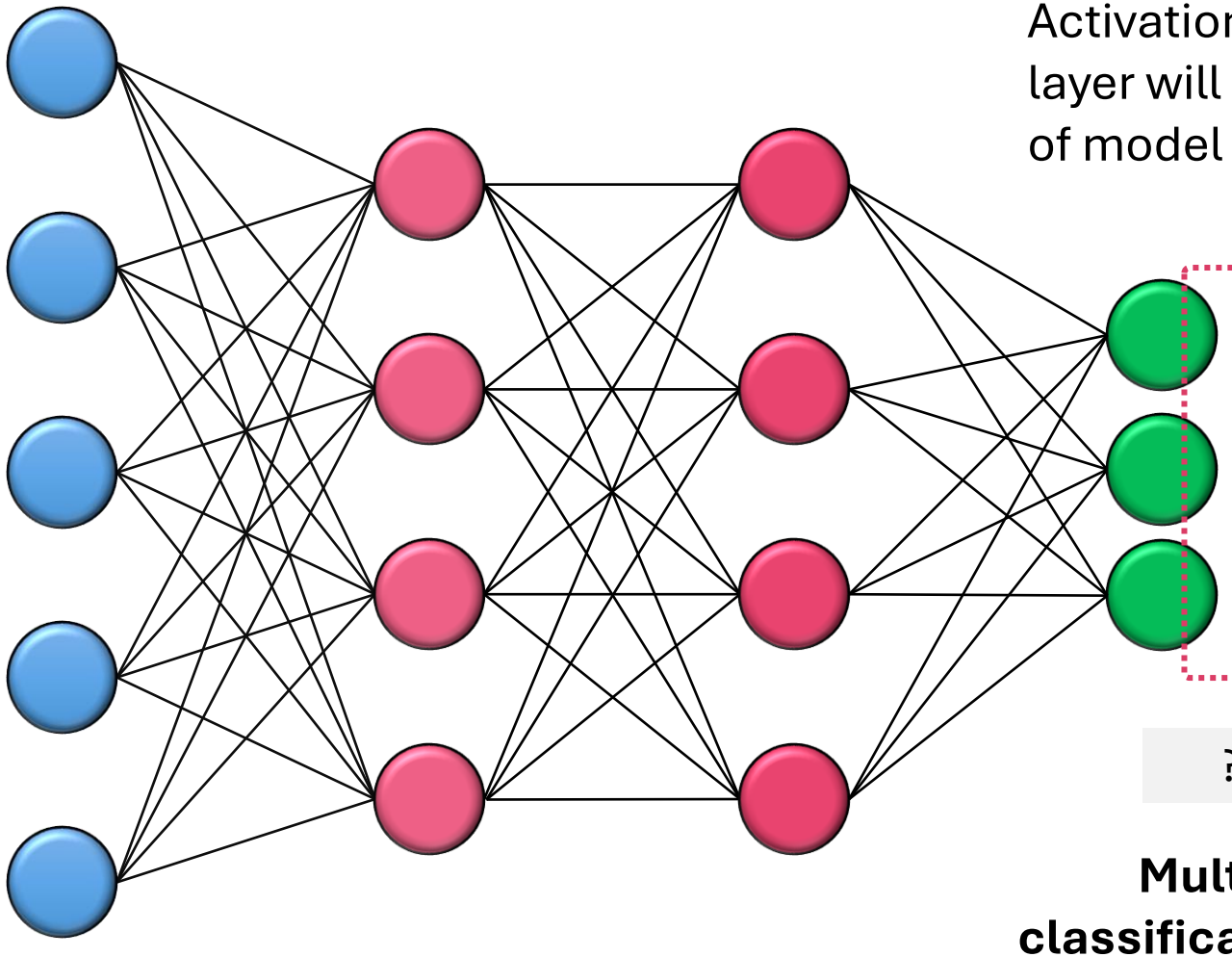
Activation function for the output layer will then define which type of model we want

```
torch.nn.
Functional.softmax()
```

Multiclass classification model

$\Rightarrow$ probabilities [0; 1] (sum = 1)

How to build a model class with PyTorch

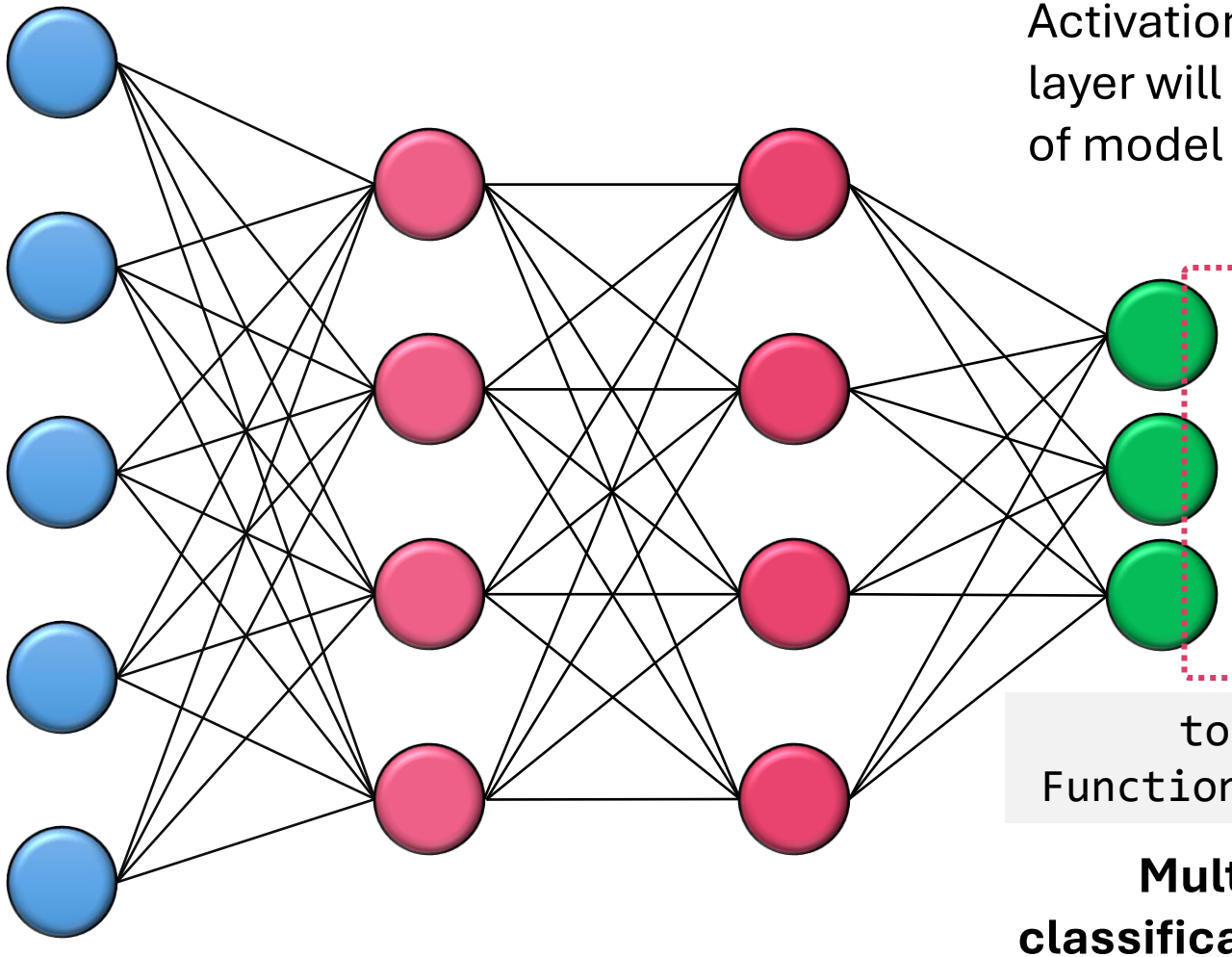


Activation function for the output layer will then define which type of model we want

??

Multilabel classification model

How to build a model class with PyTorch



Activation function for the output layer will then define which type of model we want

```
torch.nn.
Functional.sigmoid()
```

Multilabel classification model

$\Rightarrow$ probabilities [0; 1] (each independent)

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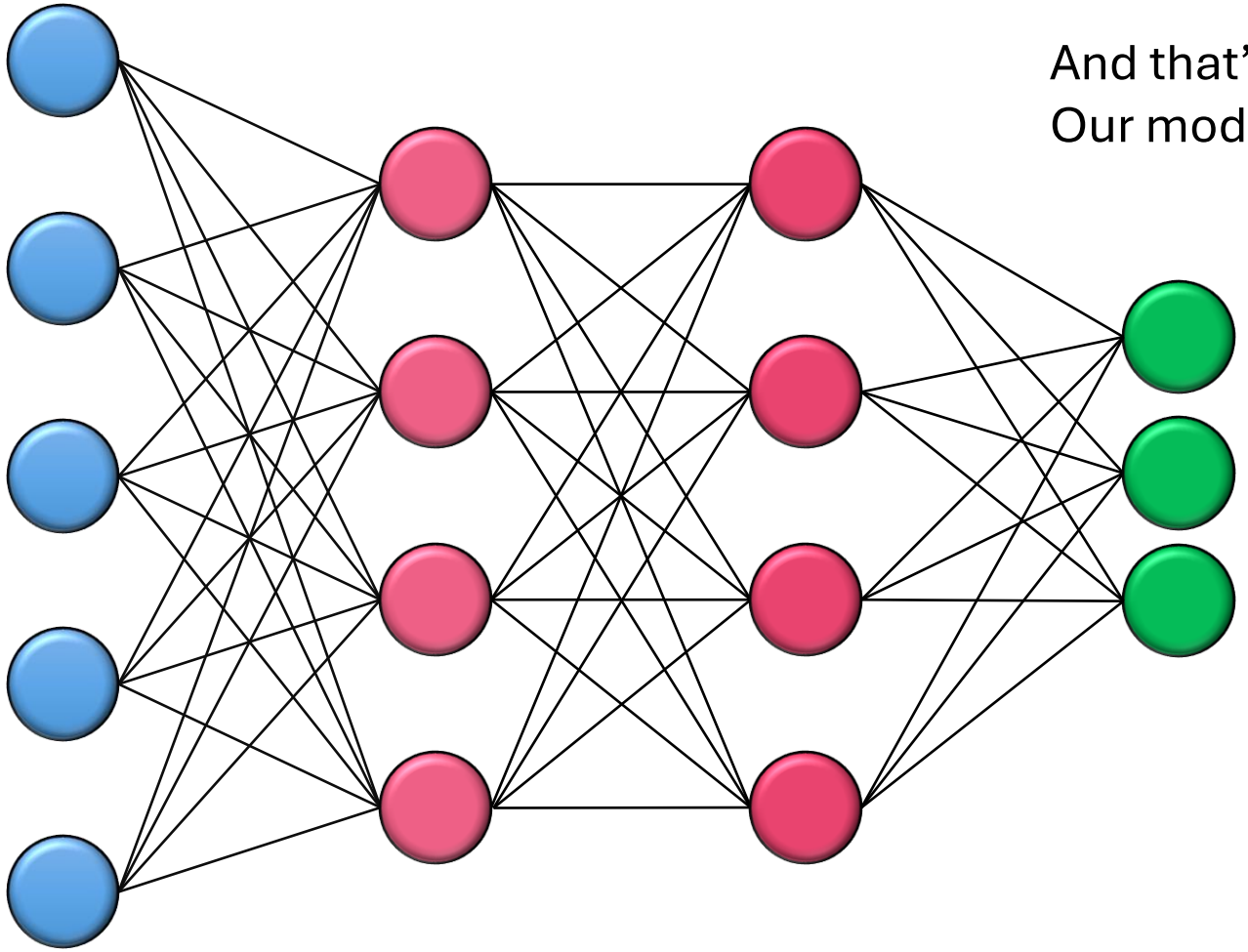
How to build a model class with PyTorch



And that's pretty all!
Our model is designed

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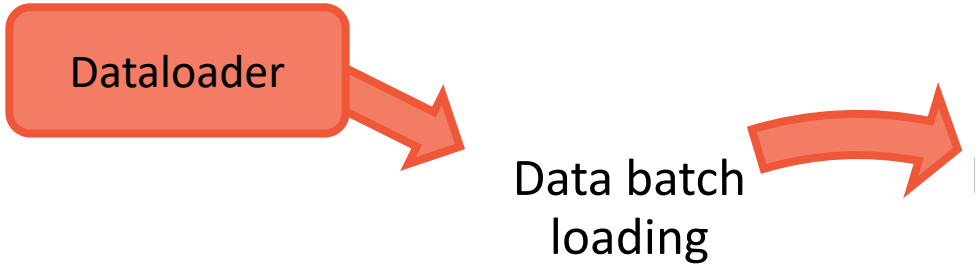
PyTorch training loop



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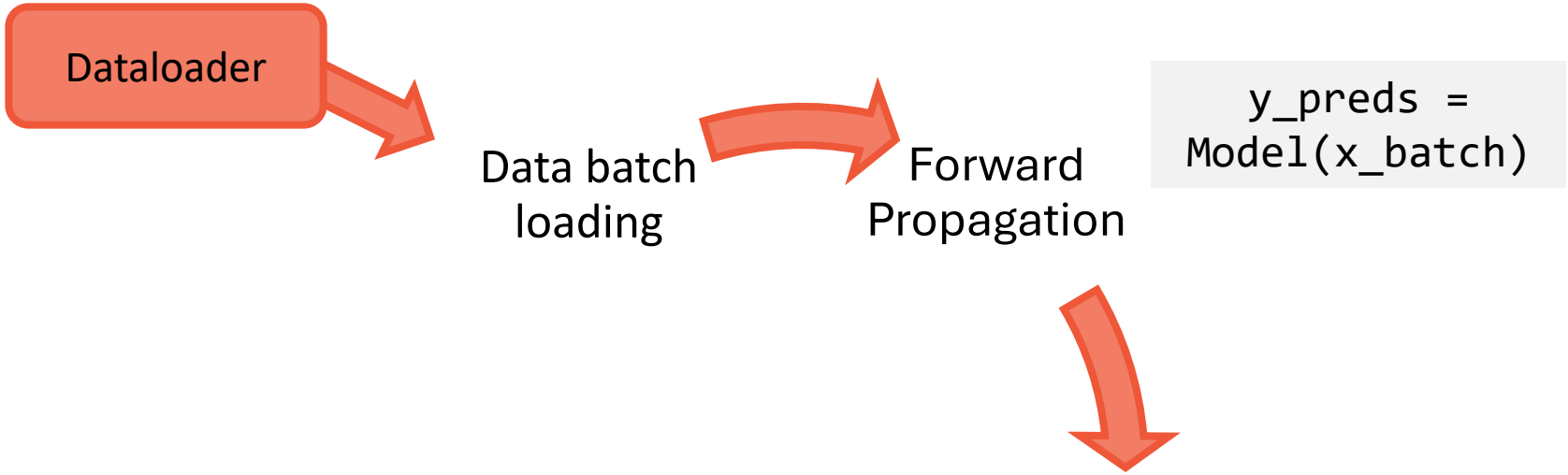
PyTorch training loop



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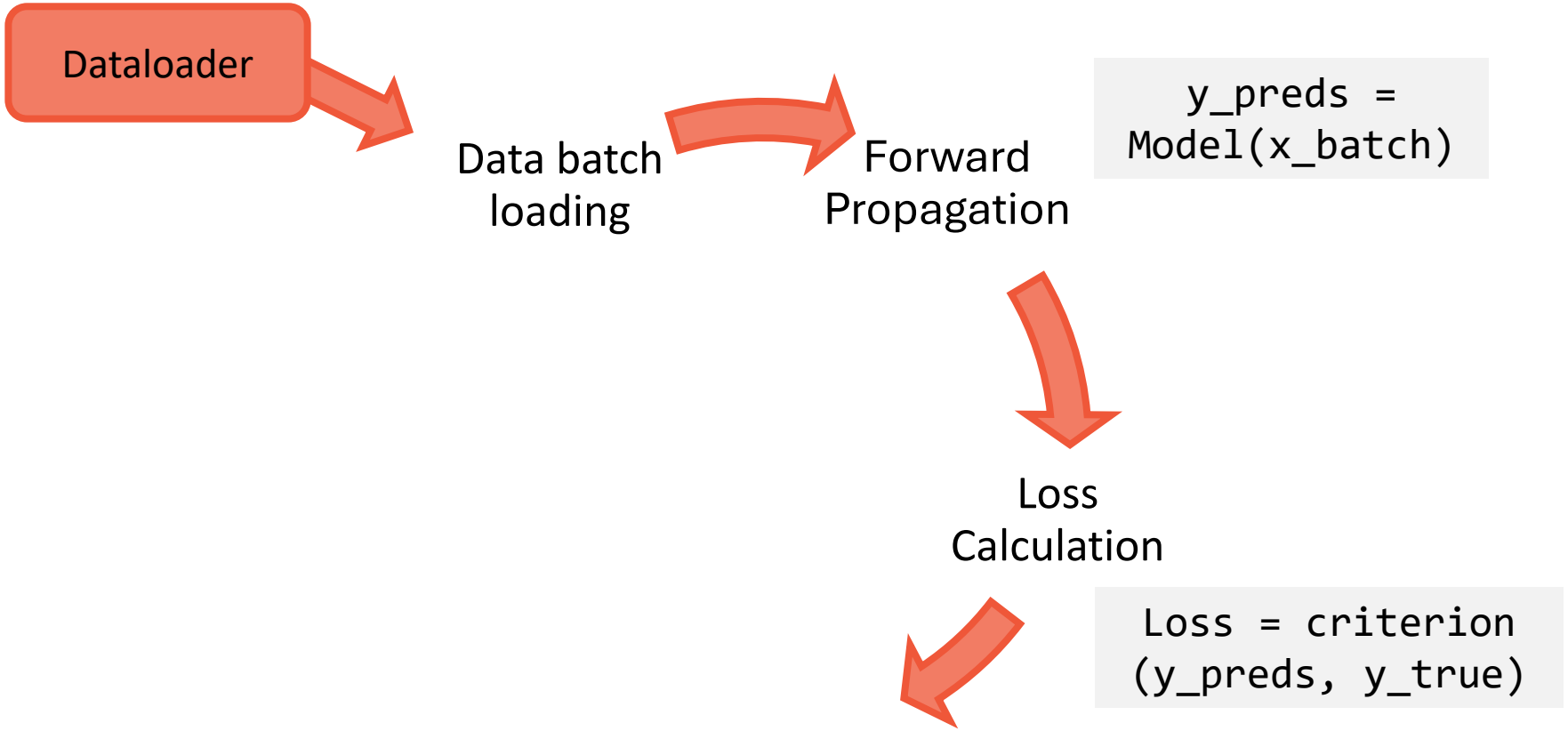
PyTorch training loop



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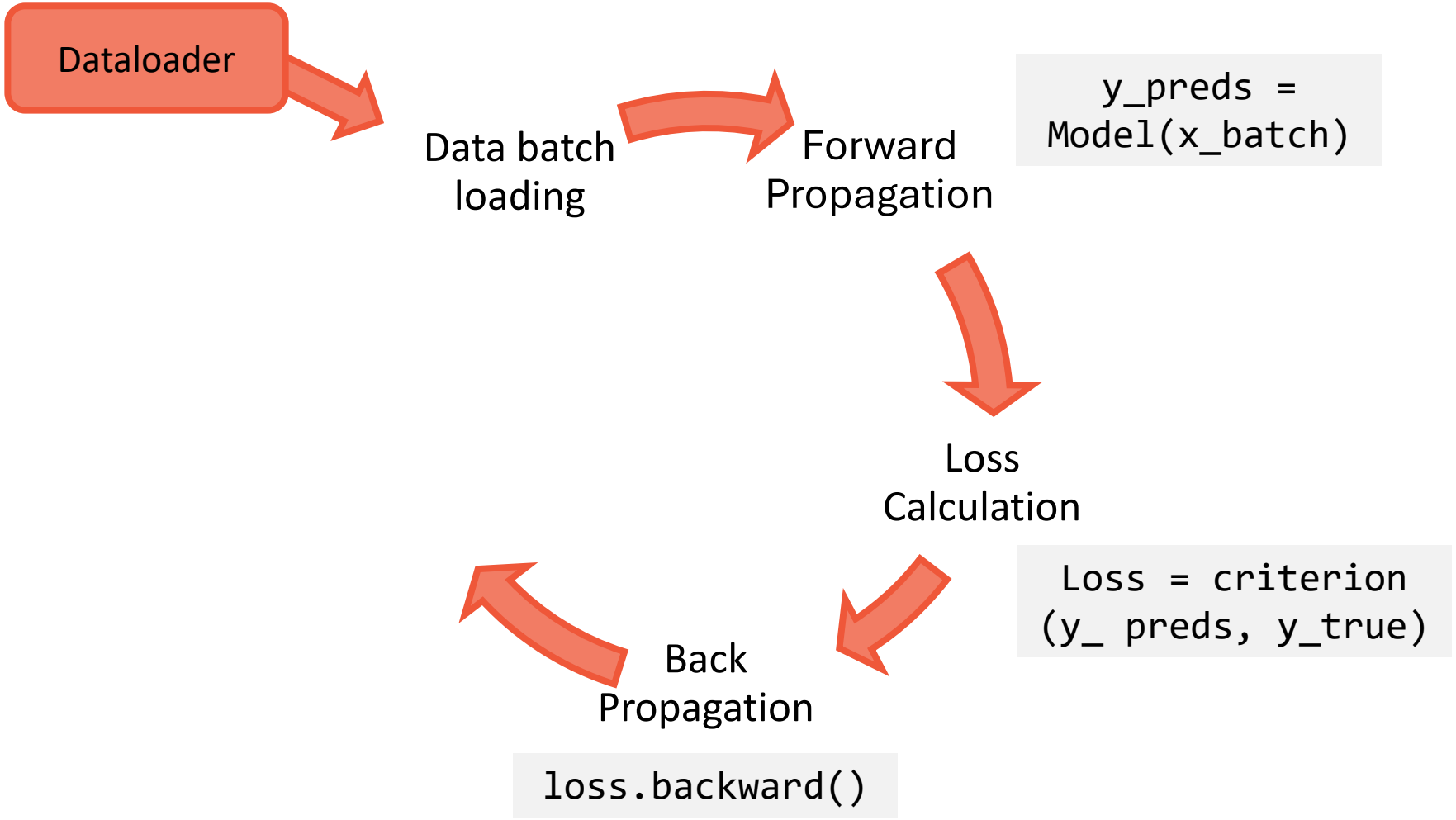
PyTorch training loop



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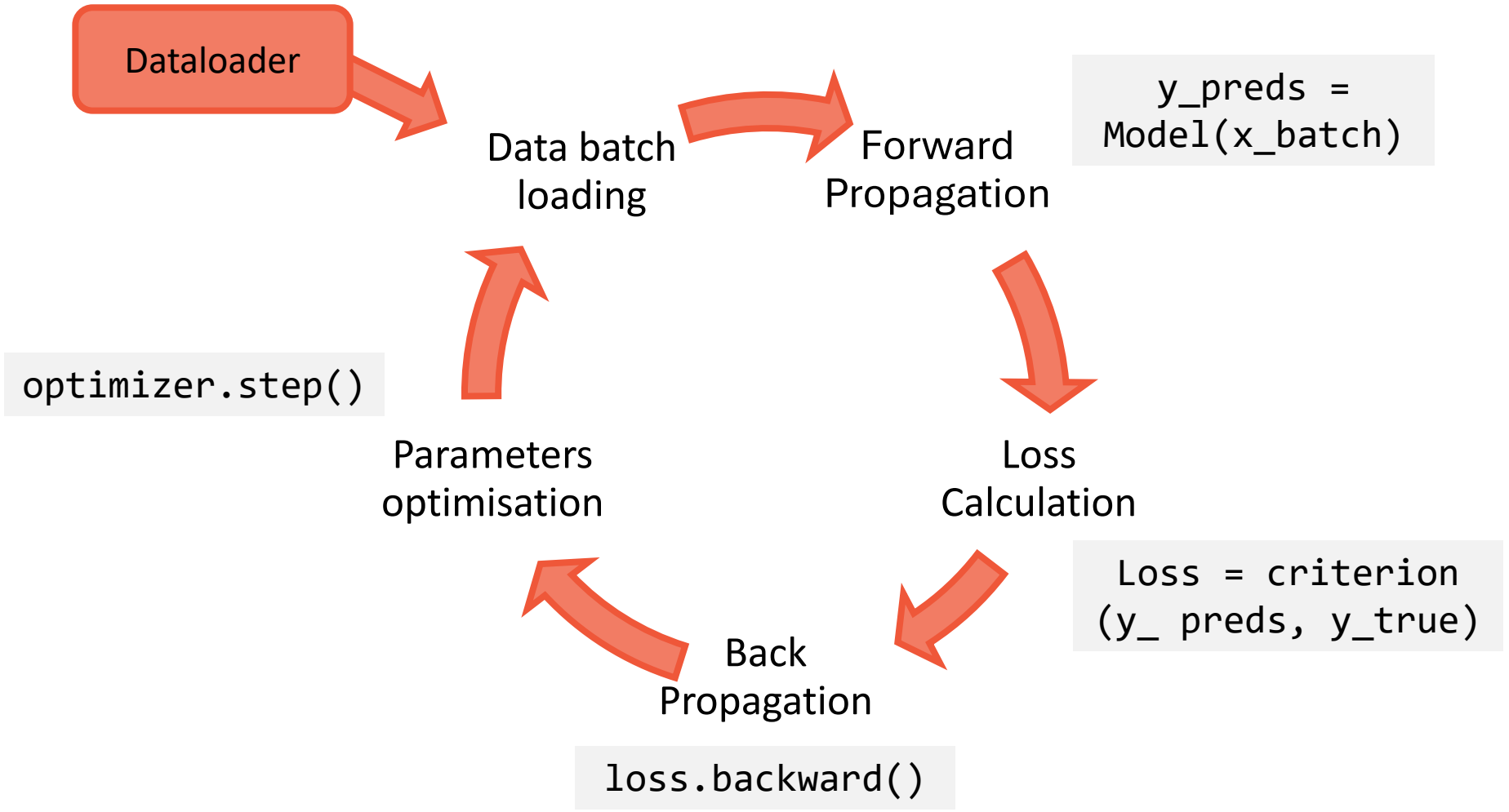
PyTorch training loop



Model design in PyTorch: binary and multilabel classifications

- Be careful!
- In PyTorch, `nn.BCEWithLogitsLoss()` internally applies both:
  - Sigmoid, to convert logits to probabilities
  - Loss calculation
- Thus, the model itself should not contain a sigmoid activation during training !
- Sigmoid activation should be used only for inference


```
# training
criterion = nn.BCEWithLogitsLoss()
logits = model(X)
loss = criterion(logits, y)
```

```
# Inference
with torch.no_grad():
    logits = model(X)
    probs = torch.sigmoid(logits, dim=1)
    preds = (probs >= 0.5).int()
```

Model design in PyTorch: multiclass classification

- Be careful!
- In PyTorch, `nn.CrossEntropyLoss()` internally applies both:
  - Softmax, to convert logits to probabilities
  - Negative log-likelihood, to compute the loss
- Thus, the model itself should not contain a softmax layer during training !
- Softmax layer should be used only for inference


```
# training
criterion = nn.CrossEntropyLoss()
logits = model(X)
loss = criterion(logits, y)
```

```
# Inference
with torch.no_grad():
    logits = model(X)
    probs = torch.softmax(logits, dim=1)
    preds = torch.argmax(probs, dim=1)
```

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